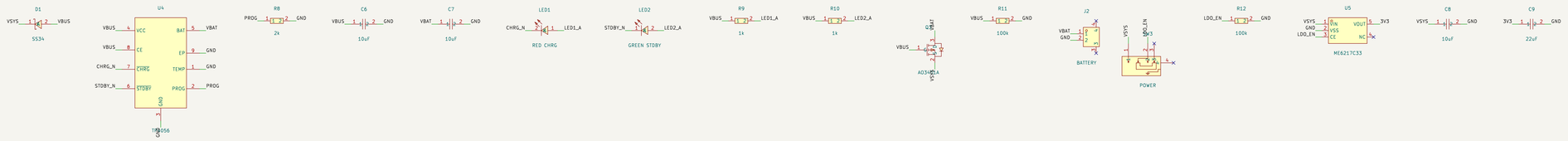
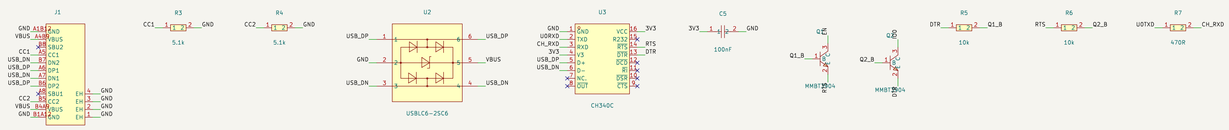
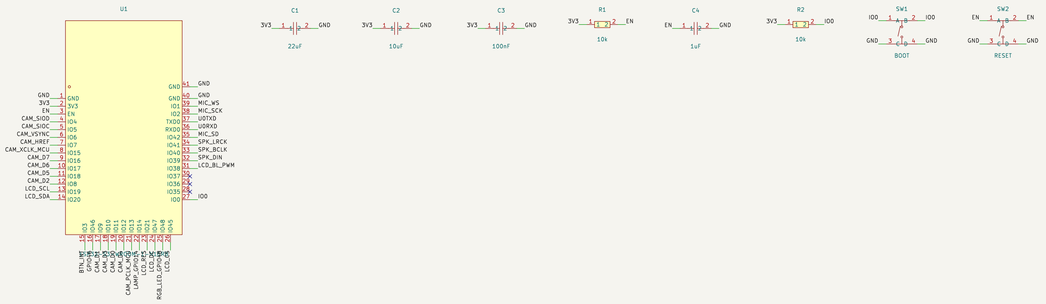
POWER: USB-C 5V / TP4056 charger / power-path / 3V3 LDO



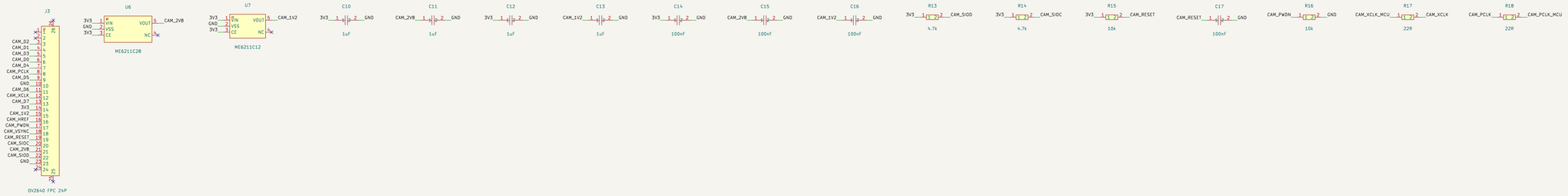
USB-UART CH340C + auto-download (DevKitC style)



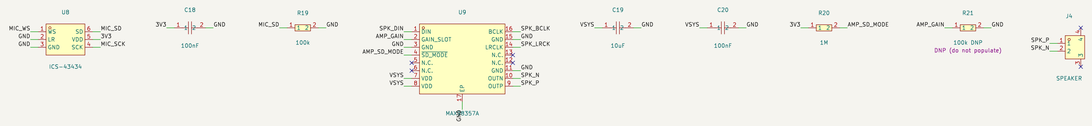
ESP32-S3-WROOM-1-N16R8 core



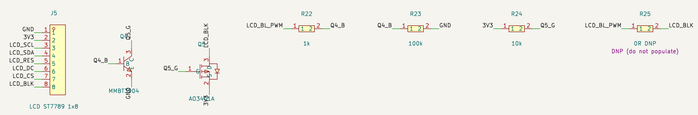
CAMERA OV2640 24P DVP (xiaozhi bread-compact-wifi-s3cam)



AUDIO: ICS-43434 I2S mic + MAX98357A I2S amp



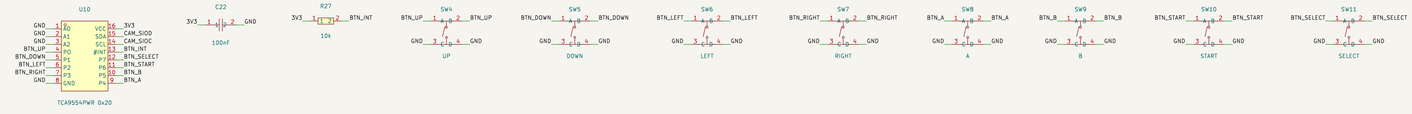
LCD ST7789 240x320 SPI + backlight



WS2812 status LED + expansion header



GAME BUTTONS (v2): TCA9554 @0x20 on SCCB bus (GPIO4/5), INT->GPIO3, D-pad + A/B/START/SELECT



NOTES: (1) GPIO19/20 are used by LCD -> native USB unavailable; flash via CH340C (SW3 must be ON).

(2) GPIO45 (LCD_CS) is a strapping pin: no pull-up on this net, check LCD module.

(3) J3 FPC pin order = ESP32-CAM style (pin1=Y0). Verify against your OV2640 module + connector contact side.

(4) R21, R25, J6 = DNP. Battery J2 polarity: pin1 = +, verify your battery lead.

(5) v2: U10 TCA9554 I2C addr 0x20 shares camera SCCB (GPIO4 SDA / GPIO5 SCL). INT -> GPIO3 (R27 pull-up). Official xiaozhi firmware ignores the buttons.

Symbols/footprints: LCSC/EasyEDA via easyeda2kicad. GPIO45 = LCD CS: NO pull-up (strapping).
Auto-generated from BOM/pin map. NOT built or tested. Check every footprint & FPC orientation.

Maker Bu Wan Ji Hua

Sheet: /

File: ESP32S3_AI_Chatbot.kicad_sch

Title: ESP32-S3 AI Chatbot (xiaozhi bread-compact-wifi-s3cam pinout)

Size: A1	Date: 2026-10-06	Rev: 0.2 (v2 game buttons)
KiCad E.D.A. 9.0.2		Id: 1/1